

Finite-Contact Audit of a Selected Graphene Transmission Target

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Abstract—We freeze an ideal massless-Dirac strip model to audit a selected lowest-mode transmission target 7/768. Infinite-mass boundaries, identical finite-doping contacts, neutrality in the strip and abrupt mode-preserving interfaces give explicit current-normalized transmission. A 50 meV lead supports no mode at the infinite-contact design width. Finite contacts at 200 meV reduce lowest-mode transmission by about 11 percent. Numerical refitting restores the target within this model, but a one-percent width perturbation shifts transmission by about six percent. These results establish model consistency and sensitivity, not material realization, an occupancy trigger or a dissipative latch coupling.

Index Terms—Graphene transport, contact matching, Dirac strip, mode transmission, interface calibration.

I. FROZEN MODEL AND TARGET PROVENANCE

The selected target is $T_* = 7/768$. It is a transmission design choice inspired by a historical UD capacity parameter; equality of these numbers does not equate transmission probability with occupancy, backlog or completion. No graphene material invariant is inferred.

Following the ideal strip setting of [1], impose infinite-mass transverse boundaries and zero interior chemical potential. The longitudinal length is $L = 30$ nm and the declared continuum parameter is $\hbar v_F = 0.6582119$ eV nm. Identical leads share the same transverse confinement, with chemical-potential energy $\mu_\ell > 0$ relative to their local Dirac point. Interfaces are abrupt and preserve mode index. Temperature, disorder, interactions and magnetic field are zero. This is a mathematical boundary model, not a demonstration that hBN or lithography supplies these conditions.

The protocol fixes lead energies 0.05, 0.10, 0.20, 0.50, 1.0 eV, a nominal finite-contact design at 0.20 eV, and width sensitivity of $\pm 1\%$. These settings are declared before this successor evaluation. The target was exposed previously; inversion is engineering tuning, not an independent prediction.

Modes have $q_n = (n + 1/2)\pi/W$, $n \geq 0$. A propagating lead requires $q_n < k_F = \mu_\ell/(\hbar v_F)$, with $k_x = \sqrt{k_F^2 - q_n^2}$. Modes at or beyond threshold are not incoming scattering channels. Spin and valley multiplicities are not included in the reported mode sums; physical conductance requires a separate degeneracy and contact contract.

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TABLE I
FINITE LEADS AT FIXED W_∞ ; MODE COUNTS EXCLUDE DEGENERACY

μ_ℓ (eV)	Open modes	T_0	Relative change
0.05	0	—	—
0.10	1	0.005080882	−44.26%
0.20	1	0.008109228	−11.03%
0.50	4	0.008953863	−1.76%
1.00	7	0.009074408	−0.44%

II. FINITE-CONTACT MATCHING

For a single mode, use $H/\hbar v_F = -i\sigma_x \partial_x + q\sigma_y$ inside the neutral strip. Its transfer matrix is $P = \text{diag}(e^{qL}, e^{-qL})$. Incoming and reflected lead spinors are

$$u_+ = \begin{pmatrix} 1 \\ (k_x + iq)/k_F \end{pmatrix}, \quad u_- = \begin{pmatrix} 1 \\ (-k_x + iq)/k_F \end{pmatrix}. \quad (1)$$

Continuity gives $P(u_+ + ru_-) = tu_+$. Symbolic solution, with identical incoming/outgoing current normalization, yields

$$t = \frac{1}{\cosh(qL) + i(q/k_x) \sinh(qL)}, \quad (2)$$

$$T_n = \frac{1}{\cosh^2(q_n L) + (q_n/k_{x,n})^2 \sinh^2(q_n L)}. \quad (3)$$

The sign of the amplitude phase depends on transfer conventions; the probability is unchanged. Six scoped exact checks verify the symbolic amplitude, modulus, current factor and bookkeeping identities. Twenty-seven binary64 checks compare direct spinor matching, flux conservation and the fitted root, with tolerance 10^{-12} . Floating checks are not interval-certified dimensional fabrication bounds.

For $k_F \rightarrow \infty$, $T_n \rightarrow \text{sech}^2(q_n L)$, recovering the ideal-contact result [1]. At finite doping, transmission approaches zero when the longitudinal lead wavevector vanishes. An exponential attenuation alone omits interface matching; even the ideal-contact large- qL asymptotic form includes prefactor four.

III. CONTACT AND MULTIMODE RESULTS

The ideal-contact lowest-mode inversion is

$$W_\infty = \frac{\pi L}{2 \text{arcosh} \sqrt{768/7}} \approx 15.502297 \text{ nm}. \quad (4)$$

Its lowest transverse energy is approximately 66.694429 meV. Therefore a 50 meV uniform-width lead has no propagating mode at this width. A contact with different width or confinement needs a new matching problem; its modes cannot be silently substituted. At 0.20 eV, refitting the lowest-mode target gives $W \approx 15.795235$ nm. Two modes are then

TABLE II
DECLARED SENSITIVITY AROUND THE FINITE-CONTACT FITTED WIDTH

Width change	T_0	Relative target change
-1%	0.008562755	-6.05%
0	0.009114583	0 (fit)
+1%	0.009689538	+6.31%

open. The fitted lowest-mode transmission is $7/768$ within binary64 root tolerance, while the mode sum is approximately 0.009114585752 , exceeding the target by $2.42 \cdot 10^{-9}$. Thus a per-mode probability and the trace $\text{Tr}(t^\dagger t)$ remain distinct even when the additional contribution is small.

Table II records the declared width perturbations.

This is a deterministic parameter sensitivity, not a distribution of fabrication errors or an empirical confidence interval. It does not include edge-family changes, which would alter the frozen boundary model itself.

IV. IMPLEMENTATION BOUNDARY AND NEXT GATE

No gate voltage is inferred: the dielectric geometry alone does not establish a constant lever arm, local potential or gapped-ribbon density of states. No material scattering limit

is derived. The chosen transmission neither detects the $7/768$ occupancy threshold nor produces a Lindblad rate. Both transduction steps remain OPEN.

The next useful test is finite gate-detuning and energy-window transport under this same contact model, followed by physically motivated boundaries and disorder. Acceptance must concern a declared tolerance band, including mode sums and phase behavior, rather than exact rational equality from a fitted scalar. Thermal averaging and contact losses must be included before any measured-performance claim. The finite-contact model is an adjacent interface study; native admission, autonomous completion and physical latch compilation remain separate open gates. Physical promotion is zero.

The replay preserves the frozen protocol, symbolic checks, numerical receipts, dependency versions and hashes. Independent review and physical calibration are pending. No quantum advantage, noise screening, fault tolerance or device readiness follows from these calculations.

REFERENCES

- [1] J. Tworzydło, B. Trauzettel, M. Titov, A. Rycerz, and C. W. J. Beenakker, "Sub-Poissonian Shot Noise in Graphene," *Phys. Rev. Lett.*, vol.96, 246802, 2006. Author preprint: arXiv:cond-mat/0603315v3.